Technische Universiteit Eindhoven University of Technology

Student name:

Student number:

Examination cover sheet

(to be completed by the examiner)

Course name: Particle-based simulations Course code: 6EMA02

Date: 31 October 2018

Start time: 13:30 End time: 16:30

Number of pages: 5

Number of questions: 4

Maximum number of points/distribution of points over questions: 25-25-25-25

Method of determining final grade: Grade written exam: total points divided by 10. The grade for the written exam has to be at least 5.0.

The final grade for the course is computed as: 50% assignment + 50% written exam.

Exam inspection: E.A.J.F. Peters

Other remarks: You are allowed to answer questions in English or Dutch. Sometimes you are asked to use an earlier result in a subsequent question. If you do not have this result assume a value and explicitly state the assumption.

Instructions for students and invigilators

Permitted examination aids (to be supplied by students):

This is a closed book exam. So no material other than those stated below are allowed

□ (Graphical) calculator

□ Dictionar(y)(ies). If yes, please specify:any bilangual dictionary

Important:

- examinees are only permitted to visit the toilets under supervision
- it is not permitted to leave the examination room within 15 minutes of the start and within the final 15 minutes of the examination, unless stated otherwise
- examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- the house rules must be observed during the examination
- the instructions of examiners and invigilators must be followed
- no pencil cases are permitted on desks
- examinees are not permitted to share examination aids or lend them to each other

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, etc.
- using a clicker that does not belong to you
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- visiting the toilet (or going outside) without permission or supervision

Section 1. General aspects of particle-based simulations (25 points, 5 each)

a) Describe the role of Brownian motion and dissipative forces in, respectively, microscopic, mesoscopic and macroscopic particle-based simulation methods.

b) In a simulation particles 1 and 2 have a non-bonded interaction described by the Buckingham potential:

$$\varphi(r) = A\, e^{-B\, r} - \frac{C}{r^6}\,, \text{where } r = |\mathbf{r}_1 - \mathbf{r}_2|,$$

with A, B, C constants. Derive the expression for the x-component of the corresponding force experienced by particle 1.

c) What is a conservative force?

d) In a particle-based simulation you have pair-forces that are zero beyond a cutoff length, r_{cut}. In a force computation subroutine, when considering all particle-pairs in the system, the computational effort scales as N^2, where N is the number of particles in the system. Describe a method to reduce the computational effort to order N instead of N^2.

e) Mathematically show that, for updating positions in a particle-based simulation, the Verlet time-integration scheme is higher order than the Euler-forward scheme.

Section 2. Microscopic Simulations (25 points, 5 each)

To determine transport coefficient of methane you want to perform atomistically detailed molecular dynamics (MD) simulations of pure methane at various temperatures. The system is a periodic box of dimensions $10.8\ \text{nm} \times 10.8\ \text{nm} \times 10.8\ \text{nm}$ filled with 20000 molecules, where the hydrogen atoms are explicitly accounted for. You want to perform equilibrium measurements in an NVT ensemble.

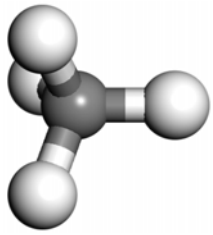


Figure 1: Atomistically detailed methane consisting of one C and four H atoms.

a) Name the relevant bonded and non-bonded interactions in this system and describe how these interactions are modelled in an MD simulation.

b) What determines the time step used in the MD simulation? What do you expect to be its value expressed in SI units? (A rough estimate is sufficient.)

c) What would be the average kinetic energy of the system at 103 K? **Hint: You can use the Boltzmann constant from the table in Section 3.**

d) Explain in detail how you would impose the temperature in the MD simulation.

e) You want to compute the shear-viscosity, η, using the Green-Kubo relation:

$$\eta = \frac{V}{k_B T} \int_0^\infty \langle \tau_{xy}(0)\, \tau_{xy}(t) \rangle \, dt \,, \text{with}$$

$$\tau_{xy} = -\frac{1}{V} \sum_{i=1}^{N} \left(m_i \, v_{x,i} v_{y,i} + x_i \, F_{y,i} \right)$$

Here τ_{xy} is the microscopic expression of the shear stress of the system with volume V, and for particle i: m_i: mass, $v_{x,i}$, $v_{y,i}$: x and y components of velocities, x_i: x-coordinate of positions and $F_{y,i}$: y-component of total force acting on the particle.

Describe how you would implement the computation of the shear viscosity using this relation in an MD code.

Section 3. Mesoscopic Simulations (25 points, 5 each)

Blood consists of 2 primary parts: blood plasma that acts as the fluid and red blood cells that are colloid particles. A novel tool to diagnose diseases uses a micro channel filled with blood. To estimate the concentration of the red blood cells, the red blood cells are separated from the plasma using a centrifuge (similar to the sedimentation problem discussed in class).

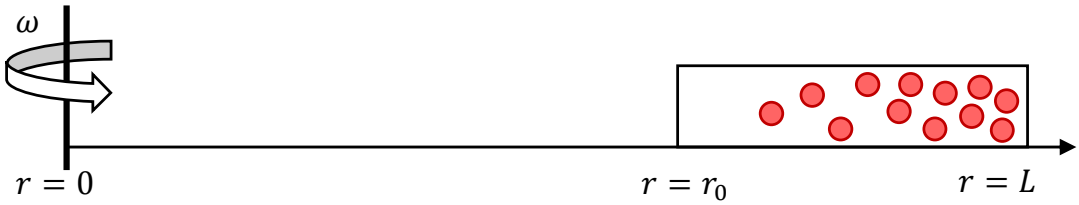


Figure 2: The channel rotates with an angular velocity ω about a fixed axis located at r=0. At $r = L$ the end of the channel is located. The length of the channel is $L - r_0$.

You can model the system in a co-rotating reference frame. This means that the effect of rotation is incorporated by a centrifugal force, which can be incorporated as a 'gravity' that acts in the r-direction and is r-dependent. The corresponding 'gravitational acceleration' is $g_r = \omega^2 r$. Other fictitious forces that can arise due to rotation, such as the Coriolis force, are negligible.

system data	symbol	value (SI units)
mass density blood plasma	ρ_p	$1025\ \mathrm{kg\ m^{-3}}$
viscosity blood plasma	μ_p	$1.30 \times 10^{-3}\ \mathrm{Pa\ s}$
mass density red blood cells	ρ_c	$1150\ \mathrm{kg\ m^{-3}}$
radius red blood cell	R_c	$3.00\ \mu\mathrm{m}$
temperature	T	$300\ \mathrm{K}$
length channel	$L - r_0$	$10\ \mathrm{mm} = 0.01\mathrm{m}$
maximum distance	L	$0.100\ \mathrm{m}$
angular velocity	ω	$250\ \mathrm{rad\ s^{-1}}$
constants		
Boltzmann constant	k_B	$1.38064852 \times 10^{-23}\ \mathrm{m^2\ kg\ s^{-2} K^{-1}}$

a) You are considering to model the separation of the red blood cells using a mesoscopic simulation method. Summarize the main features of: 1) Langevin Dynamics, 2) Dissipative Particle Dynamics, 3) Stochastic Rotation Dynamics.

b) Use a force balance on a single red blood cell to derive an expression for its terminal velocity at $r = L$. For this estimate of the terminal velocity assume that red blood cells can be approximated as spheres, that flow of the plasma is in the Stokes regime and that a no-slip boundary condition for the fluid velocity holds on the red blood cell wall.

c) Compute the most relevant dimensionless numbers: Pe and Re.

d) Based on the result of c) discuss if a mesoscopic particle-based simulation method is suitable to model this system. If you could not answer c) discuss two cases: 1) $Pe = 30$, $Re \ll 1$ and 2) $Pe = 10000$, $Re \ll 1$.

e) What specific simulation method, mesoscopic or otherwise, would you advice to use to simulate the separation phenomenon? Motivate your answer.

Section 4. Macroscopic Simulations (25 points, 5 each)

You want to investigate the flow of a fluid through a cylindrical packed bed of spherical particles. An important step is creating the packed bed by means of simulating the filling of the tube using a discrete element method (DEM) simulation.

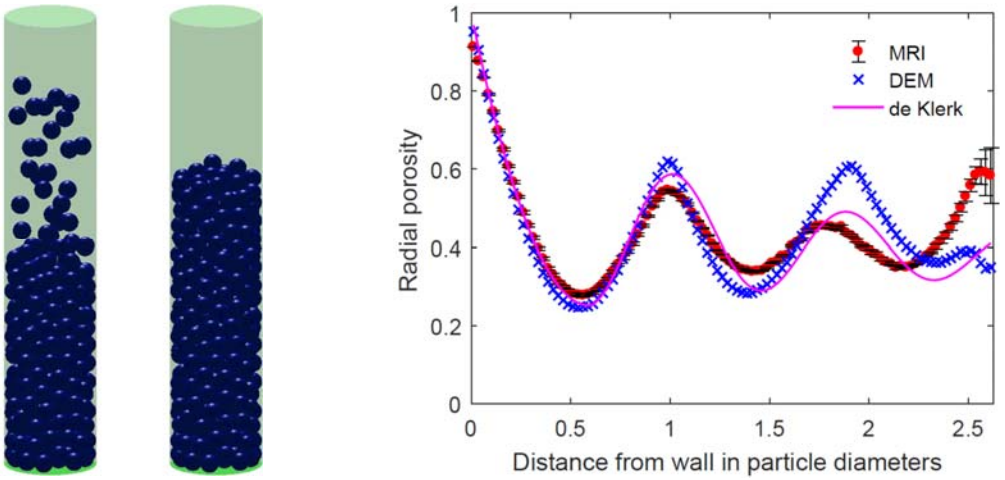


Figure 3: DEM simulation snapshots of filling a tube with spherical particles (left). Porosity profile near the tube wall in comparison with theoretical predictions (by de Klerk) and MRI measurements (right).

a) To model the particle-particle and the particle-wall interactions you use the soft-sphere model. What are the 5 key parameters of this model?

b) Describe in words how you would include the cylindrical tube in your DEM simulation.

c) Fluid flow in a slender packed bed exhibits by-passing near the wall because here the local porosity is larger. To quantify this phenomenon you would like to compute a radial porosity profile as shown Figure 3 (right). Describe how you would compute such a profile?

d) Next, you would like to compute the flow of water through the packed column. What computational method would you use for this? Summarize the main features of the chosen method in maximally five sentences.

e) How would you experimentally validate the computed water flow?